

Hamdard - A Maternity Companion in Urdu

Kaavish Report
presented to the academic faculty
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In partial fulfillment of the requirements for
Bachelor of Science
Computer Science

Dhanani School of Science and Engineering

Habib University
Spring 2021

Hamdard - A Maternity Companion in Urdu

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Dedication

For our parents who remained supportive, patient and encouraging to us throughout the four years of our journey.

For all women out there who have been subject to body-shame and low self-esteem due to social taboos and cultural discrimination.

Acknowledgements

First and foremost, we would like to extend our praises and gratitude to God, the Almighty, for His blessings throughout our final year project, for opening pathways for us when we felt challenged and for guiding us in completing our FYP successfully.

We would like to express our deepest gratitude to our supervisor, Dr. Syeda Saleha Raza, for giving us the opportunity to complete our FYP under her guidance. Her guidance and support were truly invaluable for us. Her sincerity and motivation inspired us, and her innovative ideas and dedication towards our project brought us clarity and helped us achieve our project goals. It was a great privilege and honor to work and study under her guidance.

We also extend our gratitude to Team Rehnuma, for inspiring our project and giving us the initial data to use. Hamdard would not have been possible without Rehnuma.

We are extremely grateful to our parents and other family members for their constant love, support and prayers, for the sacrifices they endured for giving us a good education and supporting our dreams and career ambitions.

We would also like to thank the Kaavish Working Group, who were receptive to our queries, and worked with such commitment to make our final year project journeys as smooth as possible, especially in such uncertain times of the pandemic.

All of our professors, who we have interacted with during the four years of our journey at Habib university. It has been an honor for us to have studied under the guidance and mentorship of such qualified and knowledgeable professors. They have truly shaped us into who we are today.

We would like to extend thanks to the Habib community, that instills a sense of social responsibility, as well as the push to achieve excellence while remaining empathetic and ethically conscientious. All of our friends and colleagues, who have supported us when things became challenging and made this journey exciting.

Lastly, we would like to extend our thanks to Dr. Shanza, Dr. Momal and Ms. Abeer Mian for taking out the time from their busy schedules to help us in understanding the maternity domain, and all the women who have helped us in

any way to bring our project to fruition, whether by filling out the survey form, recording empathetic messages or testing our conversational agent. These women are our inspiration, and we hope to make lives of mothers easier through our maternity companion.

Abstract

Hamdard is a contextual AI assistant built on RASA framework and deployed on a web application. It acts as a personal companion to pregnant women, making use of Natural Language Processing (NLP) to answer their maternity related questions in the Urdu language, predicting complications through Bayesian predictive modeling, as well as providing psychological and emotional assistance. Hence, Hamdard aims at providing women multiple levels of support throughout their nine-months of pregnancy.

Contents

1	Introduction	12
1.1	Problem Statement	12
1.2	Proposed Solution	12
1.3	Intended User	13
1.4	Project gantt chart and deliverables	13
1.4.1	Deliverables	13
1.4.2	Gantt Chart	14
1.5	Key Challenges	16
2	Literature Review	17
2.1	Chatbots for Healthcare	17
2.2	Chatbots to Assist Pregnant Women	19
2.3	RASA and NLP Chatbots	20
2.4	Chatbots Involving Bayesian Networks and PL bots	21
2.5	Chatbots in Urdu Language	21
2.6	Conversational Assistants in Non-English languages	22
3	Software Requirement Specification (SRS)	25
3.1	Functional Requirements	25
3.1.1	<u>Login</u>	25
3.1.2	<u>Complication Prediction</u>	26
3.1.3	<u>Weekly messaging</u>	26
3.1.4	<u>Typing in queries</u>	27
3.1.5	<u>User sign-up</u>	27
3.1.6	<u>Maintaining chat history</u>	28
3.1.7	<u>Identifying and confirming spelling mistakes</u>	28
3.1.8	<u>Keeping track of context</u>	29
3.2	Non-functional Requirements	29

3.3	External Interfaces	30
3.3.1	User Interfaces	30
3.3.2	Application Program Interface (API)	35
3.3.3	Hardware/Communication Interfaces	35
3.4	Use Cases	36
3.5	Datasets	37
3.6	System Diagram	40
4	Software Design Specification (SDS)	41
4.1	Software Design	41
4.2	Data Design	42
4.2.1	WhatsApp Messaging functionality and Personal Dashboard	43
4.2.2	FAQ conversational functionality	43
5	Development	45
5.1	Web App Development	45
5.2	Chatbot Development	45
5.2.1	Conversational Data Preparation	46
5.2.2	Transliteration	48
5.2.3	Urdu Language Processing	48
5.2.4	Programming Custom Actions	48
5.3	Complication Prediction	48
5.3.1	Identifying Complication and Symptoms	48
5.3.2	Integrating Bayesian Network in RASA	51
5.4	Weekly Messages	53
5.4.1	Automating Running the Script	53
5.4.2	Querying the Database	53
5.4.3	Twilio API	53
6	Experiments and Results	54
6.1	Talking to Hamdard	54
6.2	Data Validation	55
6.3	Testing Results	55
6.3.1	Testing within RASA	55
6.3.2	Testing Custom Actions	62
6.3.3	Testing using sample User Statements from Survey	62
7	Conclusion and Future Work	65

Appendix A Data	66
Appendix B Code	67
References	68

List of Figures

1.1	Gantt Chart Kaavish 1	15
1.2	Gantt Chart Kaavish 2	16
3.1	Home screen	30
3.2	User login screen	31
3.3	User dashboard screen	32
3.4	User registration screen (for users to subscribe for weekly messaging and complication prediction services)	33
3.5	Weekly messaging service to users on WhatsApp	34
3.6	Use-Case Diagram	36
3.7	A snapshot of the FAQ data-set	37
3.8	A snapshot of the Hospital location data-set	38
3.9	A snapshot of the weekly messaging data-set	39
3.10	System Diagram	40
4.1	UML Class Diagram for the system	41
4.2	A snapshot of the weekly messaging data-set	42
5.1	Training examples stored in nlu.yml	46
5.2	A snapshot of stories	47
5.3	Complication Prediction Disclaimer	49
5.4	Complication Predictor Bayesian Network	51
5.5	Intents to trigger Complication Predictor	52
6.1	Testing Chatbot	54
6.2	Testing Complications Predictor	55
6.3	Intent Confidence through Test/train data represented as a histogram	57
6.4	Intent Confidence through Test/train data in a confusion matrix	58
6.5	Intent Confidence through Cross Validation represented as a histogram	60

6.6	Intent Confidence through Cross Validation in a confusion matrix . .	61
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List of Tables

5.1	Complications and Symptoms	50
6.1	User Input Testing	63
6.2	Breakdown of User Input Testing	64

1. Introduction

1.1 Problem Statement

Due to the conservative setting of Pakistani families, females are not allowed to learn or discuss their health issues. Talking openly about health issues regarding pregnancy, menstruation and other women's issues is seen to be taboo in Pakistan, due to which women suffer physically and mentally. All women don't have the privilege to go and address every single question they may have during their nine-month pregnancy period with a doctor either, since that is not free of cost. Due to this, the majority of women remain unaware of the mental and physical health issues relevant to them. A 5-year study conducted by the Agha Khan University's Psychiatry Department showed that out of the patients receiving psychotherapy, 65% were women. Most of them were married where the distress with spouse/in-laws triggered their psychiatric illness [19]. This is just one example among many in our society which highlights the crucial need for a solution that brings reliable health opinions on the fingertips of women. Our project aims to present that effective solution.

1.2 Proposed Solution

The aim of this project is to create a system which would provide maternity related medical and emotional assistance to the Urdu speaking women. The main features of this assistant are:

- Converse with the user in Urdu language, answer their questions and predict complications during pregnancy.
- Empathetic support through WhatsApp messages.

1.3 Intended User

Hamdard aims to provide support and maternity-related information to women who are pregnant or are planning on getting pregnant soon, to give them advice on how to look after themselves and their baby during their pregnancy. Through conversations with this maternity partner, women will be able to know what to expect during pregnancy and labour which will help them make educated decisions about their well-being. Since pregnancy is a time of uncertainty, Hamdard will also provide emotional support through its messaging functionality. Husbands and family members are also encouraged to use Hamdard’s services so that they can know more about pregnancy and how it affects the mother. The assistant would give answers and respond to questions in Urdu, hence the user should know how to read and type in Urdu.

1.4 Project gantt chart and deliverables

1.4.1 Deliverables

1.4.1.1 Kaavish 1

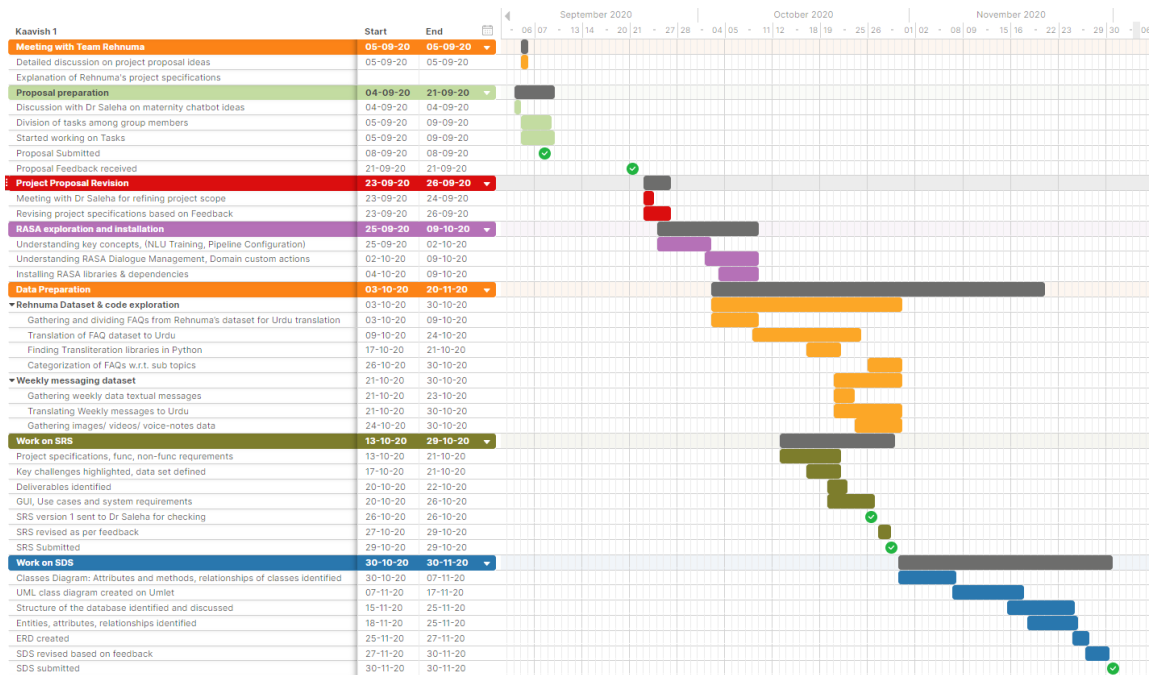
1. SRS, SDS and Literature Review documents.
2. Data-set in implementation form i.e., categorized into intents, stories, entities format.
3. Application prototype-Bot is able to give appropriate responses to queries in Urdu and convert Romanized queries into Urdu and answer them. (20-30% of the chatbot)
4. Rudimentary implementation of weekly messaging. The prototype would be something like the following: (a) On typing in the week number, the week’s messages and resources would be received. (b) WhatsApp integration needs to be completed to a certain level.

1.4.1.2 Kaavish 2

1. Complete Project Report
2. Complete web application with chatbot widget that answers queries in Urdu language as well as transliterates Romanized Urdu queries into Urdu and provides appropriate responses.

3. Complication Predictor integrated with the chatbot, asking user for symptoms and providing appropriate predictions.
4. Messaging functionality-pre-compiled messages for useful information and emotional support sent weekly on user's WhatsApp number upon registration.

1.4.2 Gantt Chart



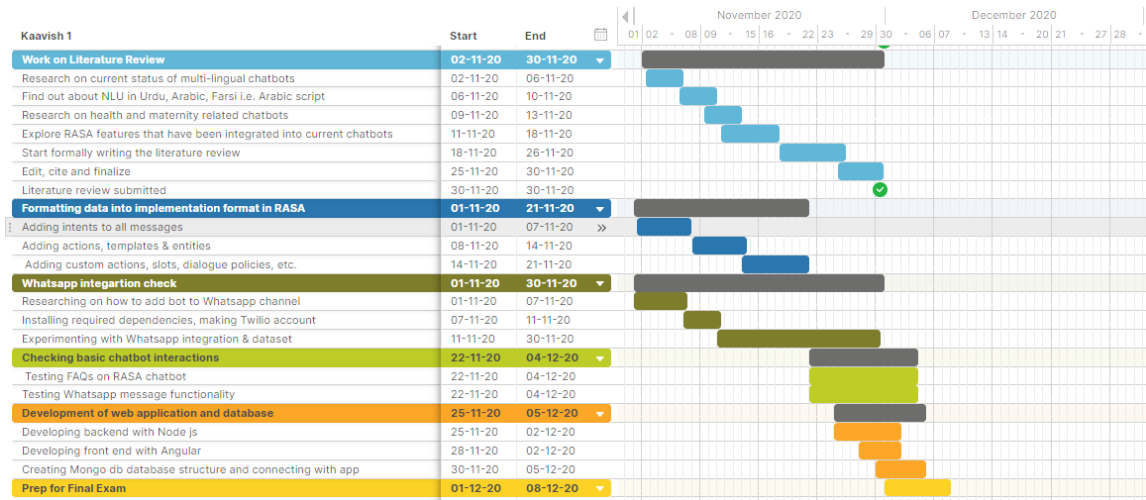


Figure 1.1: Gantt Chart Kaavish 1

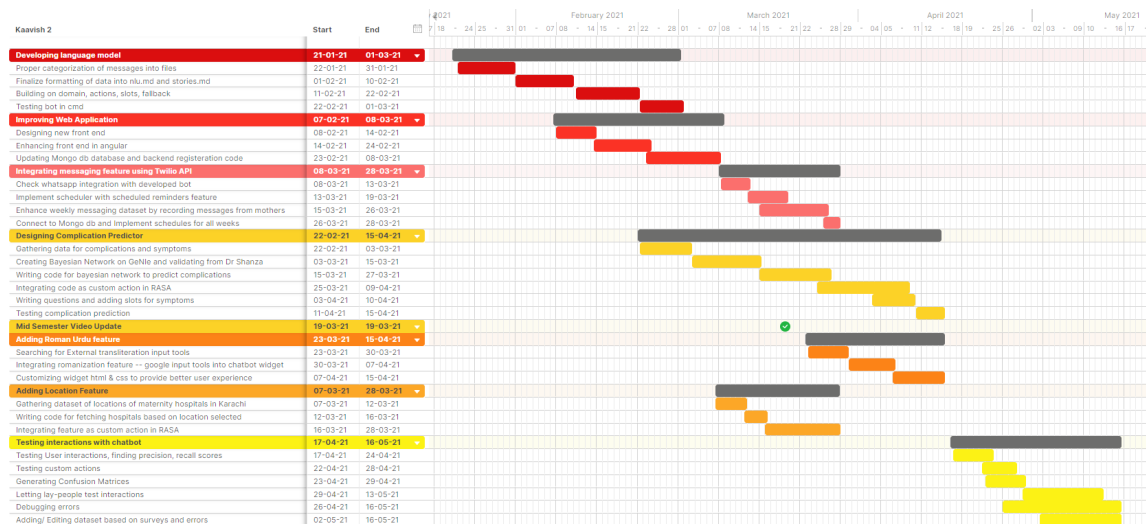




Figure 1.2: Gantt Chart Kaavish 2

1.5 Key Challenges

One of the challenges we came across was the translation and transliteration of our data set. It was challenging to find a tool which could translate our huge data set (Rehnuma’s existing data) into Urdu and Roman Urdu for NLP to be applied on it in RASA Framework. We overcame this challenge by using Google Transliteration Library in Python and other online resources to reach our target efficiently.

Another challenge faced was during the collection, verification and addition of the data set for common pregnancy complications and the integration of Bayesian Network to RASA framework for predicting complications based on user symptoms. It was difficult deciding which pregnancy complications and symptoms to leave out, and which to include in our data. For this, we explored data sets available online, and consulted a domain expert to verify symptoms and complications and suggest changes accordingly.

Another challenge we faced was how to make the medical language easily understandable for common man and even illiterate people. We revised our messages after getting valuable suggestions from our domain expert on board.

2. Literature Review

This chapter summarizes research of previous work that is relevant to the domain of our project.

The aim of this project is to create a system which would provide maternity related medical and emotional assistance to the Urdu speaking women. The main features of this assistant are to converse with the user in Urdu language, answer their questions and predict complications during pregnancy and provide empathetic support through WhatsApp messages. Hamdard will give women advice on how to look after themselves and their baby during their pregnancy. From finding nearest maternity clinic to providing nutritional advice needed for the user's well-being, Hamdard aims to provide support to women throughout their nine-months of pregnancy. Through conversations with this maternity partner, women will be able to know what to expect during pregnancy and labour which will help them make educated decisions about their well-being. Since pregnancy is a time of uncertainty, Hamdard will also provide emotional support through its messaging functionality. Husbands and family members are also encouraged to use Hamdard's services so that they can know more about pregnancy and how it affects the mother. The assistant would give answers and respond to questions in simple Urdu so that all advice is easy to understand.

2.1 Chatbots for Healthcare

Chatbots have been a rapidly growing feature for various fields. Healthcare is one of the major fields that has made use of chatbots to assist patients efficiently. According to different statistics and data, it is found that a majority of people find it more convenient to seek help online according to their own schedule instead of paying a visit to the doctor for every single thing[21].

Be it simply answering simple questions, recommending certain treatments, helping people with first aid, or guiding them through an emergency situation, chatbots

have proven to be very useful for the healthcare sector. This section will look into several prominent examples of such chatbots assisting patients regarding healthcare around the globe. Northwell Health, one of New York’s largest healthcare services came up with an AI-based solution in the form of a conversational chatbot that plays an important role in the diagnosis of colorectal cancer, and helps patients keep track of the procedure throughout[22]. Another virtual healthcare assistant called “SafedrugBot” is not only an assistant for patients but also for doctors; it serves as a companion for women to monitor and guide them through pregnancy and breast-feeding, while also keeping the doctor updated on the dosage of drugs, so they can guide the patients accordingly[1].

According to ReferralMD, an e-consultant for healthcare: “(SafedrugBot) is capable of searching through the vast database to find out detailed information on drug guides using search queries based on active ingredient or brand name of the drug in question”[1]. Along with catering to people’s physical health issues, modern chatbots are also a means to assist a person with their psychological and mental care. “Youper” is one such conversational chatbot that monitors the mental health status of a person and provides the user with personalized meditation techniques[22]. Such conversational chatbots which aim to mimic human interactions and provide emotional support use NLU (Natural Language Understanding) and NLP (Natural Language Processing).

With the same intention of bringing healthcare to the fingertips of all people, HealthTap is a conversational assistant which not only answers patient’s queries but also connects them to a doctor from a large pool of medical experts through either chat or video calling[10].

Since this service was initially accessible mostly in the US, it’s creators decided to extend its chat functionality to Facebook as well, making it easily accessible all over the globe. (ibid) Another very well known healthcare chatbot, Florence, launched in 2017 in Dortmund DEU and has been growing and developing over time. It is a chatbot that provides the services of a nurse[8]. This personal health assistant reminds the patient to take their medicine on time, keeps track of their weight, maintains a log of their menstrual cycle and also is sensitive to the patient’s mood. Taking after the name and personality of the famous Florence Nightingale, this chatbot also suggests the closest and most relevant doctors and clinics to the users. Moreover, it is integrated with widely used apps like Facebook and Skype, to ensure maximum reach[8].

Among many other healthcare chatbots, a few more names are Izzy, Forksy, Babylon Health, OneRemission, Your.Md, Buoy Health, Ada Health, Sensely, GYANT, Infermedica, CancerChatbot, and Woebot[22][1]

2.2 Chatbots to Assist Pregnant Women

This section would solely talk about chatbots which were either made for assisting pregnant women with general queries regarding their maternity period, or those chatbots which focus on some specific aspect of pregnancy.

In Kenya, the issue of mothers from low-income households not getting access to quality assistance from public hospitals was rampant. In order to combat this challenge, a menu-based messaging service was started to answer the queries of pregnant women [11]. The problem arose when such women had urgent queries regarding some complication, but the messaging service could only take the menu-based approach as it was designed to do, which was time-taking and inconvenient. A ‘Triage Bot’ was designed that used Natural Language Processing (NLP) to store the intents of thousands of user questions[11]. NLP makes use of Machine Learning to interpret language like humans would. Using pattern recognition, an NLP-based bot learns these differences, and the level of priority is given to the intents by the Triage Bot[11]. A high level urgency question could be about pain, whereas questions about nutrition and diet are of relatively lower priority.

Another pregnancy assistant chatbot called ”Chat Crecer” was first introduced in Pilas, Buenos Aires, to accompany women during their pregnancy and the first year of the baby[12]. The chatbot also gives reminders to attend prenatal and postnatal appointments and is integrated with Facebook. This chatbot ended up winning a Webby Award because of the services it provided for Argentinian women[12].

In addition to that, MamaBot is another chatbot designed to offer assistance to women and families during pregnancy. MamaBot is also based on Machine Learning and Natural Language Processing (NLP)[16].

If we look at chatbots targeted at some specific pregnancy complications, we find a chatbot which is solely for assisting pregnant women suffering from preeclampsia; a disorder caused by hypertension, along with damaged internal organs like liver

and kidneys[23]. Pregnant women use this chatbot to educate themselves about pre-eclampsia and its risks. Data about this complication is gathered from books as well as medical professionals and patients records[23]. All this data is stored on the server side of the system to accurately provide the user with diagnostics, symptoms, predictions and prescriptions for pre-eclampsia. For building, hosting and managing this chatbot, the Flow XO platform is used[23].

2.3 RASA and NLP Chatbots

Chatbot systems have been gaining increasing popularity in the field of Natural Language Processing, Artificial Intelligence, and robot systems. Such intelligent chatbots are now being used for customer service, healthcare, or simply inquiring information about a company/website. According to various different sources, chatbots made with NLP approach are easier and faster to implement than other approaches[4].

There are different approaches to train an NLP based model; one such approach discussed by Chowdhury is using knowledge based learning for your model[7]. This is relevant to our case because we are also using the same method i.e. using predefined answers to answer questions related to maternity to train our model. This way, the question can be broken down into useful units of information and those units can then be used to train your model. Using this approach, we can tackle the problem where the same question can be phrased in many different ways but in the end, the intent triggered by that question will stay the same. This approach will make sure that different responses are not triggered by similar questions[7].

In contrast, RASA NLU is used to understand language for chatbots and AI assistants. It especially focuses on entity extraction and intent classification. RASA works on two main procedures namely RASA Core and RASA NLU. “Rasa is a library of tools from the Python programming language that is open source for building any conversational software. The goal of the creators of the Rasa chatbot Framework is to create Machine learning-based dialogue management that is easy to implement, as well as bootstrap, by initiating the stages from existing sources to create something more complex and in a more efficient way, even with minimal initial training data.”[25]

2.4 Chatbots Involving Bayesian Networks and PL bots

Bayesian networks have been used in the diagnosis of Preeclampsia during pregnancy[17]. This network is for assisting physicians in diagnosing the disease. Symptoms are inserted onto the network and the joint probability for preeclampsia is analyzed, hence reducing uncertainty in the diagnostic decision [17].

Probabilistic logic bots (PL-bots) have made use of Bayesian network models to guide businesses in determining whether or not a business rule applies in a given context. [25] The Bayesian networks encode the logical expressions for the business rule. In the domain of maternity healthcare, no other chatbot has utilized probabilistic reasoning to enhance its decision-making.

2.5 Chatbots in Urdu Language

Although there has been rapid growth in the use of chatbots all over the world, the majority of chatbots we found while researching literature were in English language. Chatbots that support different languages are covered in the section “Conversational Assistants in Non-English languages”, but chatbots in Urdu language particularly are close to none. Recently, the Telangana government in India has launched a chatbot integrated with whatsapp to provide information related to Covid-19. The chatbot features three languages; English, Telugu and Urdu. Since Urdu is the second official language of the state, the government officials felt the need to offer authentic information about Covid-19 for the Urdu speaking community[24]. However, using the chatbot ourselves, we observed that it was a menu-based chatbot; the “conversation” required the user to select an option from a given set of options and the chatbot would then provide information related to the option selected. So it’s fair to say the chatbot was not able to carry out conversations, but is good for providing information.

Apart from that, Raaji, the chatbot made for tackling menstruation myths in Pakistan, aims to offer Urdu language too in the future, but currently lacks in that area, since Urdu presents a more difficult challenge for coding than English[9].

However, UMAIR (Urdu Machine for Artificial Intelligent Recourse) is one such conversational agent in Urdu created initially to help people through the process of getting their ID cards made in Pakistan. Kaleem, O’ Shea and Crockett in their paper[18]. Urdu does not have any lexical tool (like WordNet) available for processing

the language like other western languages do. The lack of concept of spacing as word separators makes the language even more complicated for a computer to process. There being multiple sentence structures and multiple meanings for one word makes any NLP approaches infeasible[18]. What UMAIR contributes to handle these issues is that it uses the Word Order Wizard (WOW) algorithm that computes similarity between strings and matches patterns in sentences. It breaks the sentences the user inputs and stores them as units, which are then used to find a similarity score by algorithms like edit distance and finding maximal weighted bipartite graphs[18].

2.6 Conversational Assistants in Non-English languages

Conversational AI support services are being developed all around the world. Since these services have the potential to provide easy and quick guidance to billions of users in every corner of the world, they are being developed in various languages with a voice that is culturally sensitive to the audience it has been designed for. This part of the paper will look at some of the latest conversational support agents that have been developed in different parts of the world. IN 2017, KAIST (Korea Advanced Institute of Science and Technology) published a paper presenting a conversational support service for providing personalized counseling to individuals facing mental health challenges such as depression and lethargy. The service observes the mood swings of the user and provides relevant advice in the Korean language. The service uses NLP (natural language processing) and NLG (natural language generation) to make sense of the user’s questions and responses as well as to generate intelligent and appropriate dialogues[15].

Furthermore, the service can receive data in the form of videos and images and it is able to recognize the emotional state of the user. The service then responds with useful information to improve the mental health of the user, recommends various activities that would improve their mood, and also takes emergency steps for dealing with urgent cases such as suicidal impulse and panic attacks.

Another recent virtual assistant has been designed by the University of Delhi, India by the name of “Aapka Chikitsak” (Your consultant). The assistant aims to deliver telehealth by connecting patients to doctors while maintaining social distance to remain safe from covid-19[14].

“Aapka Chikitsak” is a multi-lingual conversational agent to provide free education, awareness, and guidance to patients suffering from chronic diseases; it is a virtual doctor trained extensively to converse with patients in a human-like tone. In this way, it can provide easy and fast access to people living in rural areas. The assistant uses multilingual voice application technology based on NLP to provide guidance on the most prevalent diseases in the rural areas of India. The agent keeps the location of the user in context when generating medical advice. In this way, it is able to provide the user with preventative measures, health tips, home remedies, and dietary solutions based on the location they live in.

The assistant takes user queries in the form of audio messages. It then converts the audio to text and extracts meaningful information from the text by performing natural language understanding to it. Based on the semantics of the user query, the respective intents are mapped and an appropriate response is generated in Hindi or English. The virtual assistant also has the option to be connected to a messaging platform such as WhatsApp.

The Vishwakarma Institute of Information Technology, Pune India developed a health counseling service, to have smart and natural conversations with the patient. The system contains a dataset for small-scale to more severe illnesses. In this way, the service asks for the symptoms from the patient and then makes predictions of the disease they might be suffering from and provides useful advice for curing the illness. The service is available in both English as well as code-mix language i.e. the user can type messages with both Hindi and English words. The service uses Apriori algorithms that reduce redundant chats and make the conversation appear more natural and efficient[3].

There has been significant development in chatbots and conversational assistants in the Arabic language as well. BOTTA is a female chatbot developed by Dana Abu Ali and Nizar Habash. It simulates friendly conversations in the Egyptian Arabic dialect. BOTTA is able to store relevant user information such as their name, gender and age and generate appropriate answers for them during the conversation. It uses a heuristic-based approach to fetch relevant responses from a predefined set of responses[5].

Abdullah is a Conversational Intelligent Tutoring System (CITS) that aims to teach children in their pre-teen years essential lessons related to Islam in Arabic. Abdullah asks the users a series of questions and uses a variety of media such as

images, videos, and sound effects to interact and converse with the students. Based on the student's answers, it is able to understand their knowledge level and directs the conversation appropriately[2].

LANA is also a CITS, specifically developed for autistic children to aid them in their learning of Arabic writing and teaching them topics related to science. LANA provides different learning models for different students such as visual, kinesthetic and auditory so that they are able to learn independently according to their needs and learning style[6].

3. Software Requirement Specification (SRS)

This chapter provides detailed specifications of the system under development.

3.1 Functional Requirements

This section defines the basic system behavior of the three main components that are a part of this project; Urdu chat-bot, complication predictor and messaging service.

3.1.1 Login

3.1.1.1 Introduction

Admin (always) and users (when they need to access the prediction option or change their preferences) need to login.

3.1.1.2 Inputs

They will enter in their credentials.

3.1.1.3 Processing

The app will process whether admin account or user, also whether credentials are valid or not.

3.1.1.4 Outputs

If credentials are valid, will open respective screens.

3.1.2 Complication Prediction

3.1.2.1 Introduction

Users can get predictions regarding possible complications.

3.1.2.2 Input

User will answer a series of questions about their current health status (any ache, bleeding, or problem).

3.1.2.3 Processing

All data will be analyzed with the help of bayesian network.

3.1.2.4 Output

Possible list of complications will be given along with recommending consulting a doctor.

3.1.3 Weekly messaging

3.1.3.1 Introduction

Users can receive empathetic and informational text and voice messages weekly, if they opt for it.

3.1.3.2 Inputs

User can sign up, enter in their contact details, pregnancy week, and messages receiving preferences.

3.1.3.3 Processing

The data will be stored, and messages will be sent automatically according to the user's week and preference.

3.1.3.4 Outputs

Users will receive messages as per their week.

3.1.4 Typing in queries

3.1.4.1 Introduction

User can type in their queries in Urdu, and Roman Urdu.

3.1.4.2 Inputs

User types in her query in the given chat box.

3.1.4.3 Processing

The query gets sent to the bot for analysis.

3.1.4.4 Outputs

User successfully enters in the query.

3.1.5 User sign-up

3.1.5.1 Introduction

User can sign-up for weekly messages and diagnosis option.

3.1.5.2 Inputs

Users will fill in their personal information, phone numbers, and preferences regarding the weekly messages.

3.1.5.3 Processing

The information will be stored.

3.1.5.4 Outputs

The information will be used to diagnose health issues and number for sending weekly messages.

3.1.6 Maintaining chat history

3.1.6.1 Introduction

The chatbot should maintain a conversation history.

3.1.6.2 Inputs

Chat between user and chatbot should be stored in the database.

3.1.6.3 Processing

Chatbot stores text file of conversation in back-end.

3.1.6.4 Outputs

Chat history is saved.

3.1.7 Identifying and confirming spelling mistakes

3.1.7.1 Introduction

The chatbot should be able to identify spelling mistakes and confirm the deduced sentence from the user.

3.1.7.2 Inputs

User sends a message in chat.

3.1.7.3 Processing

Chatbot matches query from existing intents and comes up with best match.

3.1.7.4 Outputs

Chatbot confirms query from user if intent does not perfectly match.

3.1.7.5 Error Handling

If no similar intent exists, chatbot asks user to try again.

3.1.8 Keeping track of context

3.1.8.1 Introduction

Chatbot should keep track of context.

3.1.8.2 Inputs

The progressing conversation with user.

3.1.8.3 Processing

Chatbot stores context of the conversation to be more efficient in responses.

3.1.8.4 Outputs

Chatbot stays on track with the conversation.

3.2 Non-functional Requirements

The following are the non-functional requirements of our system:

1. The application would work on any browser on Anroid, iOS or Windows.
2. The web application would require an active internet connection.
3. If the Internet connection gets disrupted, the user would be informed.
4. The chatbot would give prompt replies.
5. The language would be emphatic, empathetic and polite.
6. UI would be user friendly and chatbot will give relevant answers or suggestions with no grammatical errors.
7. User's confidentiality and privacy will be maintained through password protected accounts and controlled access. All communications would be encrypted, and no one will have access to anyone else's account due to accounts being password protected.
8. In case of prediction, a series of relevant questions would be asked based on user's answers until a conclusion is reached. Data integrity would be assured.

9. Patient history, personal information and phone number would be required for WhatsApp weekly messages subscription through user account. A user may choose to unsubscribe at any time.
10. The application would maintain data integrity by keeping backups of all updates to the database.

3.3 External Interfaces

3.3.1 User Interfaces

This section includes our web application screens and briefly explains them.

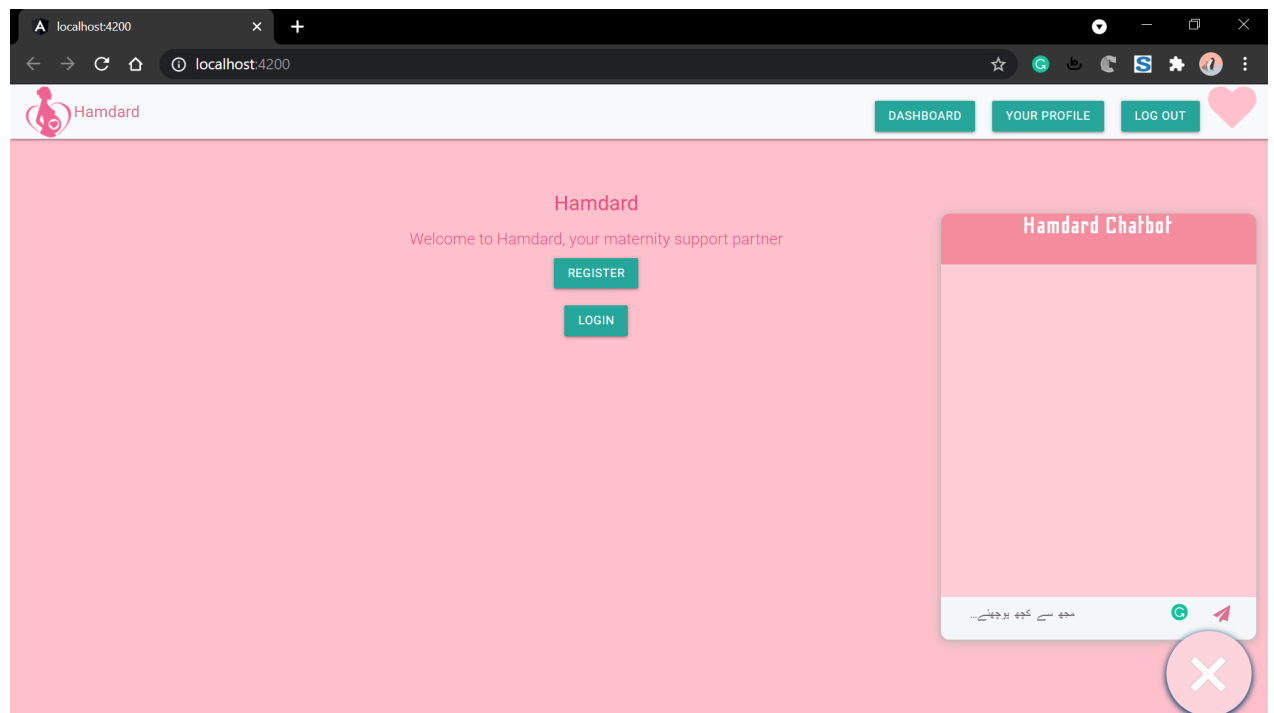


Figure 3.1: Home screen

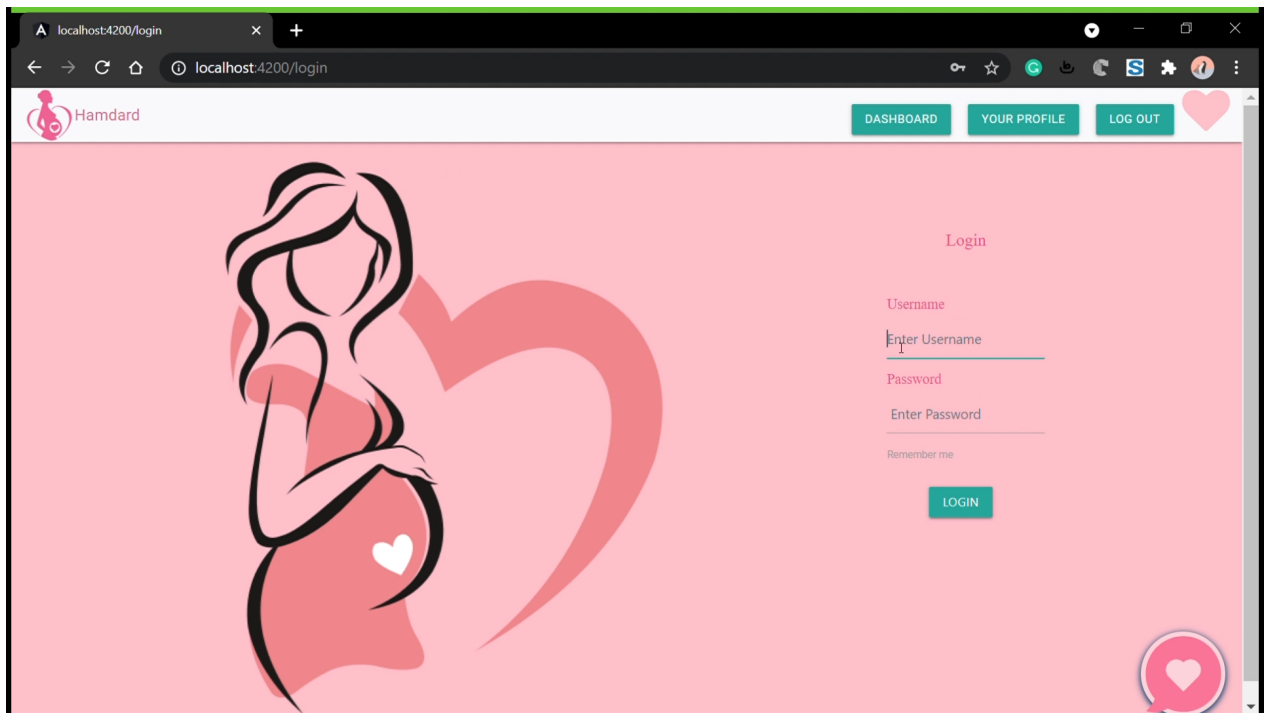


Figure 3.2: User login screen

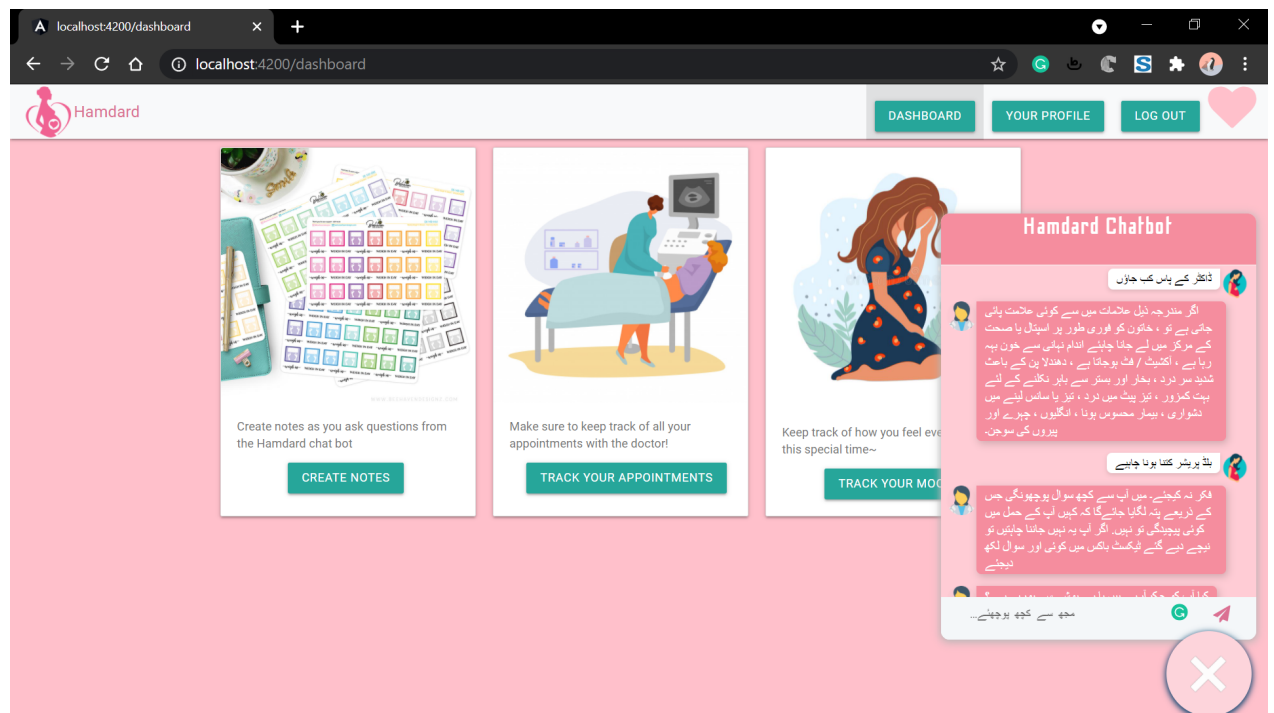


Figure 3.3: User dashboard screen

localhost:4200/register

Hamdard

DASHBOARD YOUR PROFILE LOG OUT

Register Your Account

Your Name

Enter Name

User Name

Enter Username

Phone Number

Enter Phone

We'll never share your phone with anyone else.

Pregnancy Week

21

Password

Enter Password

SUBMIT

Figure 3.4: User registration screen (for users to subscribe for weekly messaging and complication prediction services)



Figure 3.5: Weekly messaging service to users on WhatsApp

3.3.2 Application Program Interface (API)

The RASA Framework has been used to develop the chatbot. AngularJS and Node.js are used to build the web application along with Angular-Chat-Widget-Rasa library provided by Angular CLI.

For WhatsApp service, have used Twilio API.

3.3.3 Hardware/Communication Interfaces

There are no hardware interfaces required.

3.4 Use Cases

This is the complete use-case diagram for our project.

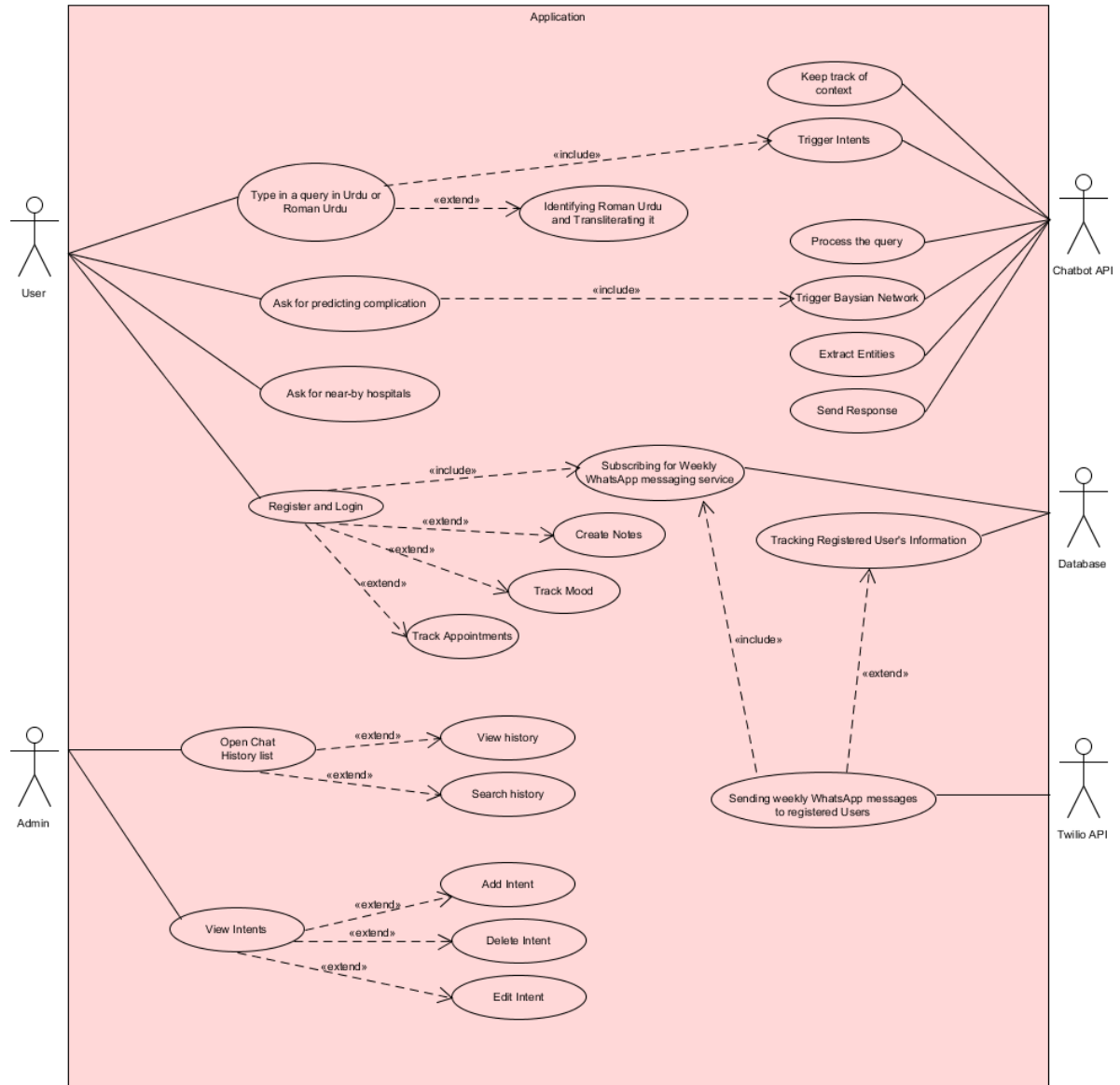


Figure 3.6: Use-Case Diagram

3.5 Datasets

The data-set consists of the following sub data-sets:

1. Maternity-related FAQs have been gathered from Rehnuma’s data-set, which were collected from surveys, local doctors and authentic websites, including WebMD [23], BabyCenter UK [4], The Government of the Bahamas’ Ministry of Health [7], Women’s Health and SurgeryCenter OBGYN [25], Women’s Health and Action Research Centre [24], and Pregnancy, Birth and Baby operated by Health direct Australia [9]. The dataset was then translated into Urdu under the guidance of public health experts and gynecologists. Medical terms were verified using NSW Ministry of Health 2012 Having a baby –Urdu. [18]

A	B	C	D	E
Intent	Question in Eng	Ques in Urdu	Answer in Eng	Answer in Urdu
antenatal_visit_attendant	Who will I be seeing at my antenatal appointments?	میں کس سے ملنے آئے گی اس کے لیے ملاقات کروائی	whether you're planning to give birth at a public hospital. If so, it's likely that you will see a doctor or midwife at the hospital.	آپ اس کے لیے ملاقات کو کس جگہ سے آپ کے لیے ایک پیشین گوئی کے لیے ضرورت ہوگی۔ اگر آپ ایک عام ہسپتال پر ولادت کی خدمات میں کرنا چاہتے ہیں، تو آپ کے لیے ہسپتال میں ایک ڈاکٹر یا میڈیکل سٹاف کے ساتھ ملنا ضروری ہوگا۔
antenatal_visit_attendant	-Who will I be seeing at my [antenatal appointments/visit_type]?	میں کس سے ملنے آئے گی اس کے لیے ملاقات کروائی		
antenatal_visit_attendant	-Who will be attending to me during my [antenatal appointments/visit_type]?	میں کس سے ملنے آئے گی اس کے لیے ملاقات کروائی		
antenatal_visit_attendant	-My attendant during [appointments/visit_type]?	میں کس سے ملنے آئے گی اس کے لیے ملاقات کروائی		
antenatal_visit_attendant	-Attendant at [hospital visits]/visit_type?	میں کس سے ملنے آئے گی اس کے لیے ملاقات کروائی		
antenatal_visit_attendant	-Who will see me during [hospital visits]/visit_type?	میں کس سے ملنے آئے گی اس کے لیے ملاقات کروائی		
antenatal_visit_attendant	-Will a nurse check me during my [visits]/visit_type or a doctor?	میں کس سے ملنے آئے گی اس کے لیے ملاقات کروائی		
antenatal_visit_attendant	-Who will check me during my [antenatal visits]/visit_type antenatal appointments?	میں کس سے ملنے آئے گی اس کے لیے ملاقات کروائی		
antenatal_visit_attendant	-Who will be seeing me during my [antenatal care]/visit_type antenatal appoinme	میں کس سے ملنے آئے گی اس کے لیے ملاقات کروائی		
antenatal_visit_attendant	-Who will be seeing me during [antenatal appointments]/visit_type?	میں کس سے ملنے آئے گی اس کے لیے ملاقات کروائی		
antenatal_visit_attendant				
antenatal_visit_attendant				
antenatal_visit_attendant				
antenatal_visit_attendant				
antenatal_visit_attendant				
antenatal_visit_attendant				
antenatal_visit_attendant				
antenatal_visit_attendant				
antenatal_visit_attendee	-Who can come along with me during my [visits]/visit_type?	میں کس سے ملنے آئے گی اس کے لیے ملاقات کروائی	It's a good idea for your birth support	ایک اچھا خیال ہے کہ آپ اپنے بچہ کی ولادت کے دوران کسی دوست یا رشتہ دار کے ساتھ ہسپتال میں جائیں۔
antenatal_visit_attendee	-Should someone accompany me for [antenatal visits]/visit_type antenatal appoinme	میں کس سے ملنے آئے گی اس کے لیے ملاقات کروائی		
antenatal_visit_attendee	-Can my [partner/attendee] come along too?	میں کس سے ملنے آئے گی اس کے لیے ملاقات کروائی		
antenatal_visit_attendee	-Can my [partner/attendee] accompany me during my [visits]/visit_type?	میں کس سے ملنے آئے گی اس کے لیے ملاقات کروائی		
antenatal_visit_attendee	-Is my [partner/attendee] allowed to come with me for [antenatal visits]/visit_type ante	میں کس سے ملنے آئے گی اس کے لیے ملاقات کروائی		
antenatal_visit_attendee	-Should I [bring my husband/attendee] alone for [visits]/visit_type?	میں کس سے ملنے آئے گی اس کے لیے ملاقات کروائی		

Figure 3.7: A snapshot of the FAQ data-set

2. For the hospital location dataset, hospitals having maternity services in Karachi were compiled using the hospital lists made by Atlas Insurance [15] and American Consulate general [14]. The contact numbers, addresses, and emails were taken from the official websites of the hospitals.

1	Address	Contact	
2	4 F 15/5 NAZIMABAD	(021) 36685560	https://aohospital.com.pk/
3	Tabish Dehlavi Road, Karachi	(021) 99260400	http://kmdc.edu.pk/abbassi-shaheed-hospital/
4	NATIONAL STADIUM ROAD	(021) 111 911 911	https://hospitals.aku.edu/Pages/default.aspx
5	515, GOLD STREET,GARDEN EAST	+92 21 32256903	
6	ATMARAM PRITAMDAS ROAD	(021) 33100007	
7	AISHA MANJIL, D – 6, BLOCK – 7,F.B. AREA	+92 21 3682296-3	
8	25/423,DARAKSHAN SOCIETY,(KALA BOARD BUS STOP),MALIR	(021) 34404040	
9	SB -9,13/C,MAIN UNIVERSITY ROAD,GULSHAN-E-IQBAL	(021) 34822261	
10	NAZIMABAD KARACHI	92-21-36616445	https://baqai.edu.pk/BaqaiHospital.php
11	TAYYAB JEE ROAD,GARIKHATA,KARACHI	(021) 32212572	https://burhanihospital.org.pk/
12	ST – 1/3,SECTOR 41 B,KORANGI TOWNSHIP	(021) 35063443	http://cgh-k.com/
13	M.A. Jinnah Road, Baba-e-Urdu Road, Nanakwara, Karachi	(021) 99215740	https://chk.gov.pk/
14	St 19 ,BLOCK -15,GULSHAN E JOHAR	(021) 111 300 999	http://dsh.com.pk/
15	METROVILLE NO. 1, S.I.T.E	021-36753407	
16	PLOT NO. 157, MALIR HALT	0305 3992945	
17	BS/3,BLOCK 4 PBS KARIMABAD F.B. AREA KARACHI	(021) 36349678	
18	A 58-59,MALIR COLONY,MAIN NATIONAL HIGHWAY	0333 2131296	
19	ST 22/B, UNIVERSITY ROAD, GULSHAN-E-IQBAL	(021) 34992706	
20	C / 42-43 ,RIZVIA SOCIETY	(021) 36608991	
21	1 DEPOY LINES M.A. JINNAH ROAD KARACHI	(021) 32258021	
22	NATIONAL STADIUM ROAD	(021) 111 456 456	https://www.lnh.edu.pk/
23	C-19, BLOCK-17, F.B. AREA, NEAR WATER PUMP	(021) 36804777	https://main.mamjihospital.com/

Figure 3.8: A snapshot of the Hospital location data-set

- Weekly messaging data consists of textual messages, relevant images and voice notes that will be used to providing important information as well as emotional support to mothers during the course of their pregnancy. These messages will be sent to the mother on her WhatsApp every week. This dataset was collected from baby bundle available on the babycenter website[4]. The voice messages were recorded from multiple women with different maternity experiences to diversify the content of the audio messages e.g., working women, old or young pregnant women etc. They can be accessed via [this link](#).

	A	B
1	Week	Messages
2	Week 1	
3	Week 2	آپ کا حمل تب شروع ہوا تھا جب آپ کا بیضہ (ovum) نطفے (sperm) سے بارآور (fertilise) ہو۔ بیضہ دو خلیات (cells) میں تقسیم ہوتا ہے۔ خلیات کی تقسیم تب تک جاری رہتی ہے جب تک کہ خلیات کی ایک چھوٹی سی گیند (ball of cells) بننے کیلئے کافی تعداد پوجائے۔ خلیات کی یہ گیند پھر فالوپین ٹیوب (fallopian tube) سے گزرتی ہوئی رحم (uterus) تک جا پہنچتی ہے جہاں یہ جھلی (lining) میں جا کر ٹھہر جاتی ہے۔
4	Week 3	اب آپ کے بچہ دانی کی جھلی (lining) میں ایک ہیٹ چھوٹی سی خلیات کی گیند (ball of cells) موجود ہو گی جو بڑھ کر آپ کے پیدا ہونے والے بچے میں ڈھل جائے گی۔ اس گیند کو، جسے بلاسٹوسائسٹ (blastocyst) کہا جاتا ہے، ایک ایسا ہارمون، ایچ سی جی (hCG) تیار کرنا شروع کرتا ہے، جو آپ کے بیضہ دانی کو بیضے یا انڈے (ovum/eggs) جاری کرنے سے روکتا ہے۔
5	Week 4	خلیوں کی گیند (ball of cells) اب ایمبریو یا جنین (embryo) میں تبدیل ہوگئی ہے۔ اب آپ اپنی گزرے ہوئے حیض (period) کے آغاز سے 4 ہفتوں کے قریب ہیں۔ یہ اس وقت کے اس پاس ہے جب آپ کا اگلا حیض (period) آنا چاہیے۔ اس وقت حمل کے ٹیسٹ میں آپ کا یازیٹو نتیجہ آں گا۔ ابھی آپ کا بچہ پوست کے بیج کے سائز کا ہے۔
		اس مہینے آپ کا حیض (period) نہیں آیا، اور آپ کے حمل کا پانچواں ہفتہ ہے۔ جلد ہی کلینک جانے کا ارادہ کریں۔
		اگر حاملہ ہو کر 12 ہفتے تک کے ساتھ شند کر رہے ہو، آپ کو آڈیٹ اور فوٹک ایسٹ کے گائڈ، دند

Figure 3.9: A snapshot of the weekly messaging data-set

Additional queries have also been added to the data-set. When the user types in their question, Hamdard will match it with the queries in the data-set and provide them with a relevant answer. Intents of each question have been identified which will be used in categorization of the FAQs and aid the assistant in finding the right answers to the queries.

3.6 System Diagram

This diagram gives a high-level view of the different components of our system and the interactions between them.

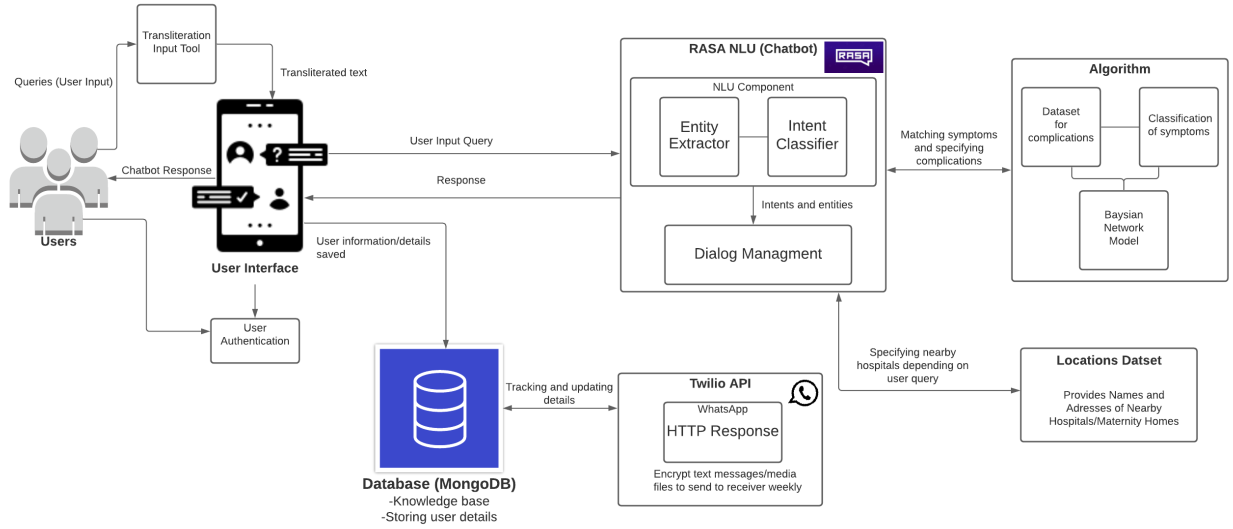


Figure 3.10: System Diagram

We have a user authentication system on our MEAN stack application and the user information is stored in the MongoDB database. The user has access to the chatbot widget where the user input is transliterated if it is given in Roman Urdu. The chatbot widget sends queries to and receives responses from the RASA NLU where intents are mapped and an appropriate response is sent. If a symptoms is sent by the user, our algorithm which uses Bayesian network is triggered, which predicts complications based on the user symptoms. The location dataset is also accessed by the RASA actions file which gives location of near-by hospitals if users ask for it. Weekly WhatsApp messages are sent to all registered users using Twilio API. The phone numbers are accessed from the database and pregnancy week is also updated.

Her phone number will be retrieved by the message scheduler from the database to send weekly WhatsApp messages using Twilio API. Notes, Appointments and Moods are further components of the Dashboard which the user can use, which are also stored in the database. The chatbot class is for the chatbot widget which will appear on user screen to access our RASA AI assistant by sending and receiving messages using REST and WebSockets for communication.

4.2 Data Design

This section presents the structure of our database that caters to persistent data storage in our project. The structure is shown as a normalized data model for relational databases. It clearly shows entities, attributes, relationships with their cardinalities, and primary and foreign keys. We have used DB designer to build our data model.

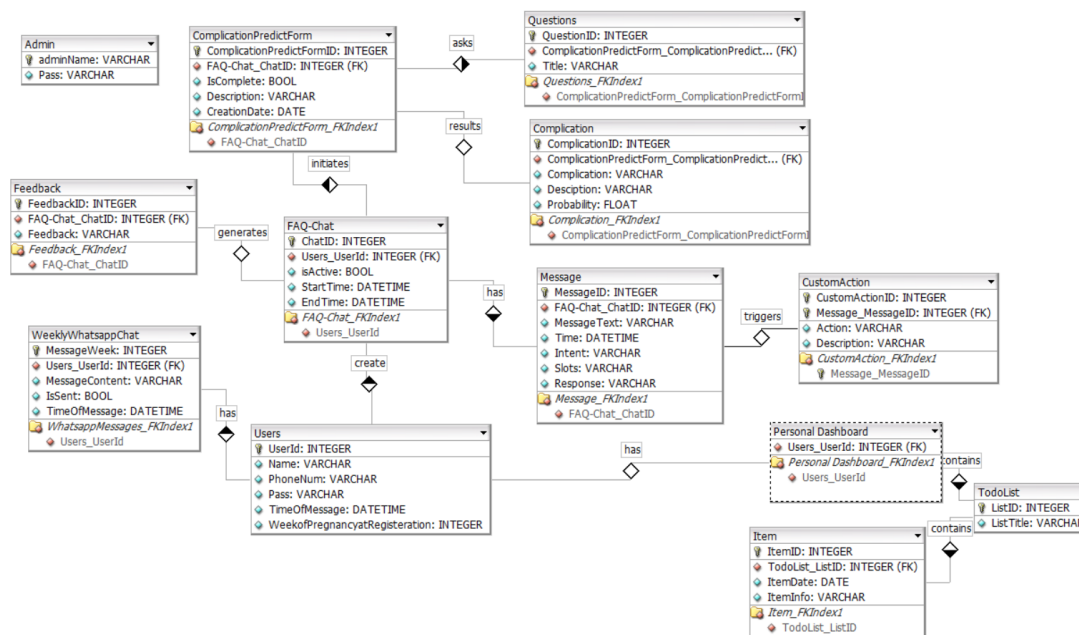


Figure 4.2: A snapshot of the weekly messaging data-set

There are two components of this project:

4.2.1 WhatsApp Messaging functionality and Personal Dashboard

These will be accessible to the users upon registration.

1. User registration: Upon registering, users will be asked to provide full name, password, current maternity week and phone number in order for them to receive weekly messages on their WhatsApp number.
2. WhatsApp Messages: Every week, the user will receive a series of helpful tips and advice according to their current week of pregnancy including text messages, voice messages, images on a specific day and time.
3. Personal Dashboard: When a user logs in using the credentials they provided while registering, they will be able to access a dashboard. This dashboard is to store their personal information and make notes on useful information they got from the chatbot. The dashboard consists of multiple lists e.g. questions to ask doctor, doctor appointments, medication, symptoms, mood, general notes etc. Each list can have multiple items added to it.
4. Admin: The admin is able to access the stored data on Mongo DB by entering their username and password.

4.2.2 FAQ conversational functionality

This will be deployed through chatbot on web application, will be accessible without registration. Each chat entity consists of FAQs by the chatbot and user, which are stored in the message entity. The message entity consists of the message from the user, the intent triggered by the user's message and the response of the chatbot. Intents and responses that will be sent by the bot upon triggering of intents are stored in the knowledge base i.e. RASA data files.

1. Chat: It is set to Active when the session is ongoing, else the isActive attribute is false. The StartTime and EndTime store the time and date for when the chat was active.
2. Feedback: Feedback for how effectively the chatbot handled each chat can be received via RASA X conversations inbox. Once a chat is complete, a summary for the messages the assistant and user has sent, as well as the predicted intents, actions and their confidences is generated. This enables the admin to best understand how users interact with the chatbot.

3. **Complication Predict Form:** Upon triggering of certain intents in a chat, the complication prediction form will be activated, where users will be asked a series of questions about their symptoms. Each of the answer to these questions will be stored in the form of slots. When all slots in the form have been filled, the predictor will generate a result with a certain probability and send the user a message about which complication they are most likely experiencing.
4. **Custom Action:** Detection of certain intents trigger custom actions. The action code runs and decides which responses need to be triggered next.

5. Development

5.1 Web App Development

With the goal of delivering an easy-to-navigate user interface, a web application using MEAN stack was developed. From client to server to database, MEAN stack refers to a collection of JavaScript based technologies used to develop web applications. The web-app offers multiple features. Users can register for their account. The form for registration includes fields for users details which will be saved at the back-end. It will be used to communicate with Twilio API and the back-end scheduler will be triggered to send weekly WhatsApp messages to the users. Users can log in to use the dashboard which helps keep track of appointments and create useful notes about health and diet. It also allows users to select mood to monitor how they feel. The chatbot widget is displayed on the homepage for the users to access without logging in or registering. It is created and customized using socket.io and REST API to communicate with the RASA server using our Angular application [13].

5.2 Chatbot Development

For the chatbot to answer pregnancy related questions, RASA framework was used. Through Natural Language Understanding, the chatbot is able to extract important entities and keywords to decipher the meaning behind the user's message. The extraction of key features helps classify the messages as specific intents, which results in the bot performing an action accordingly. The actions vary from responding to a question to sending details of the nearest maternity clinic or triggering a form that asks further questions from the user. All of this is to be defined in the Urdu model of the chatbot which processes and converts unstructured user messages into useful information, and then responds accordingly.

5.2.1 Conversational Data Preparation

- **Labelling Intents:** Once all the data for FAQs had been collected and translated, each type of question was labelled with a unique intent that complemented the meaning they carried. Every intent served as a training example of the user question for the chatbot.
- **Enhancing the training examples:** For every intent, more examples were added with alternative phrasing; synonyms, abbreviations and commonly misspelled words were also taken into account.

```
- intent: baby_due_date
examples: |
- میں اپنے بچے کی ولادت کی تاریخ کیسے معلوم کروں
- میرا بچہ کب پیدا ہوگا
- بچے کی پیدائش کی تاریخ کب ہوگی
- بچہ کون سے دن پیدا ہوگا
- ڈیلیوری کی ڈیٹ کیا ہے
- میرا حمل کب ختم ہوگا
- بچے کا جنم کب ہوگا
- بچہ کب ہوگا
- میرا بچہ کب پیدا ہوگا
- بچا کب ہوگا
- بچہ کب اس دنیا میں آئےگا
- کب ہوگا میرا بچہ
- بچے کی ولادت کب ہوگی
- بچے کے پاداش کب ہوگی
- بچہ کی پیدائش کی تاریخ بدل سکتی ہے کیا
- کیا ولادت کی تاریخ بدل سکتی ہے
- intent: baby_moves_feels
examples: |
- حمل میں بچے کی لاتیں کیسے محسوس ہوتی ہیں؟
- مجھے کیسے پتا چلے گا کہ میرا بچہ ہل رہا ہے؟
- پیٹ میں بچے کی حرکت کیسی ہوتی ہے؟
- ماں کے اندر بچے کی حرکت بتائیے
- بچہ ہلتا ہوا کیسا محسوس ہوتا ہے؟
- مجھے کیسے پتا چلے گا کہ میرا بچہ لاتیں چلا رہا ہے؟
- بچے کی لاتیں کیسی محسوس ہوتی ہیں؟
- ماں کے اندر بچے کی حرکت کیسی محسوس ہوتی ہے؟
- بچے کی حرکت حمل کے دوران کیسے تبدیل ہوتی ہے؟
- کیا میرے بچے کی حرکت وقت کے ساتھ ساتھ بدلے گی؟
- حمل کے ساتھ ساتھ کیا میرے بچے کی حرکت میں تبدیلی آئے گی؟
- بچے کی حرکت
- بچے کی لات
- بچے کی برہ رارت کیسی محسوس ہوتی ہے
- intent: baby_moves_start
examples: |
- کیا دوسری دفع حمل میں بچے کی حرکت جلدی محسوس ہوتی ہے؟
- دوسری دفع حمل میں بچے کی حرکت کب محسوس ہوتی ہے؟
- میں بچے کی حرکت پہلی بار کب محسوس کروں گی؟
- عموماً بچے ماں کے اندر حرکت کب شروع کرتے ہیں؟
- بچے کی حرکت شروع کرنے کا وقت
- بچے کی حرکت محسوس کرنے کے لئے کتنا عرصہ انتظار کرنا پڑے گا؟
- بچہ حرکت کب شروع کرتا ہے؟
```

Figure 5.1: Training examples stored in nlu.yml

- **Creating Stories:** Stories represent the actual conversation between the user

and the chatbot. For each intent, a story was created that would help define conversational flows. Picture to show the structure of stories in the data files.

```
- story: labor pain medication
  steps:
  - intent: labor_pain_medication
  - action: utter_labor_pain_medication
- story: premature labor symptoms
  steps:
  - intent: premature_labor_symptoms
  - action: utter_premature_labor_symptoms
- story: premature labor
  steps:
  - intent: premature_labor
  - action: utter_premature_labor
- story: castor oil labor
  steps:
  - intent: castor_oil_labor
  - action: utter_castor_oil_labor
- story: labor feels
  steps:
  - intent: labor_feels
  - action: utter_labor_feels
- story: active labor
  steps:
  - intent: active_labor
  - action: utter_active_labor
- story: contractions
  steps:
  - intent: contractions
  - action: utter_contractions
- story: vomit why
  steps:
  - intent: vomit_why
  - action: utter_vomit_why
- story: exhaustion
  steps:
  - intent: exhaustion
  - action: utter_exhaustion
```

Figure 5.2: A snapshot of stories

- **Maternity clinics/Hospitals Location:** In order to provide the user with information about the nearest hospitals, a script was written that indexes through an excel sheet containing information on all maternity clinics or hospitals with maternity care departments in Karachi. When a user types in the area where they live, e.g., DHA or Nazimabad, the location is saved as a slot, which is then passed to a custom action. This slot is converted into a string and searched within the excel sheet. Once found, the name of the hospital, contact details, and address are concatenated and the dispatcher utters a message containing this string.

5.2.2 Transliteration

Another requirement was the transliteration of the input text from Roman Urdu to Urdu so that the input can be processed by our RASA server. For this, an external transliteration input tool developed by Kushal Subedi on Github [20] was added. This input tool has been developed using the now deprecated Google input tools API as its back-end. If the input word is in English, it provides the transliterated Urdu text which the user can select.

5.2.3 Urdu Language Processing

To incorporate Urdu language, an extensive pipeline using the NLP Library ‘SpaCy’ was set up. Within the pipeline, the tokenizer splits up the message into individual words, which are categorized into entities and responses through the featurizer. The classifier then assigns a probability to the user message belonging to a certain intent, based on which the response selector sends the appropriate message.

5.2.4 Programming Custom Actions

For some features, custom actions had to be coded in RASA actions.py file, that would be triggered for specific intents. Examples of these intents include complication prediction and maternity hospital location selection. Action classes for these intents received entities in slots and fetched responses from the corresponding data record.

5.3 Complication Prediction

Identification of mothers with higher risk for any specific complication allows necessary precautions to be taken. Hence, an important part of our maternity companion is the complication prediction feature, which uses Bayesian Networks.

Bayesian network is considered as a standard tool in medical decision support to capture uncertainty and perform probabilistic reasoning. The Bayesian Network enabled us to intuitively establish the cause-and-effect relationship between symptoms experienced and the complication that may be diagnosed.

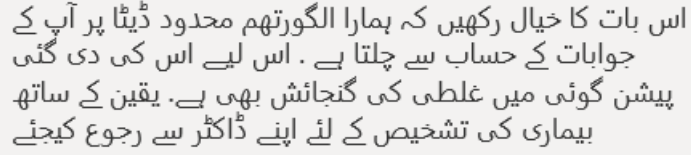
5.3.1 Identifying Complication and Symptoms

Firstly, a detailed discussion about pregnancy complications was done with domain experts. Under the guidance of Dr. Shanza Badar and Dr. Momal, the most common

pregnancy complications in Pakistan were identified along with their symptoms. A total of 8 complications were recognized and their main symptoms defined, as defined in the following table.

Once the complications and symptoms were defined, the likelihood of a certain complication being experienced with respect to each symptom was discussed, so that the conditional probability distributions can be specified that will define the causal relationships between complications and symptoms.

Since the Bayesian network has a limited amount of data to make its prediction on. Therefore, the end-result needs to be validated from the mother's gynecologist. Hence, it is extremely important for the patient to be notified about the fact that the predicted complication is not the final result, rather merely a prediction. For this, the chatbot will make the following disclaimer,



اس بات کا خیال رکھیں کہ ہمارا الگورتھم محدود ڈیٹا پر آپ کے جوابات کے حساب سے چلتا ہے . اس لیے اس کی دی گئی پیش گوئی میں غلطی کی گنجائش بھی ہے. یقین کے ساتھ بیماری کی تشخیص کے لئے اپنے ڈاکٹر سے رجوع کیجئے

Figure 5.3: Complication Prediction Disclaimer

Once all the probabilities were defined, the Bayesian Network was created on GeNIe software and shown to Dr. Shanza.

Complication	Symptoms
Anemia	Fatigue Paleness Shortness of breath Feeling faint
Ectopic Pregnancy	Abdominal pain Vaginal bleeding Feeling faint Shoulder pain
Gestational Diabetes	High blood sugar levels Extreme thirst, hunger Fatigue
Miscarriage	Vaginal bleeding Abdominal pain Vaginal Discharge
Preeclampsia	High blood pressure Headaches Swelling of hands and face Abdominal pain
Bacterial vaginosis	Vaginal Discharge with bad odour Pain or burning during urination Feeling itchy
Placenta previa	Vaginal bleeding during second or third trimester
Hyperemesis gravidarum	Nausea/vomiting Reduced appetite Feeling faint Dehydration

Table 5.1: Complications and Symptoms

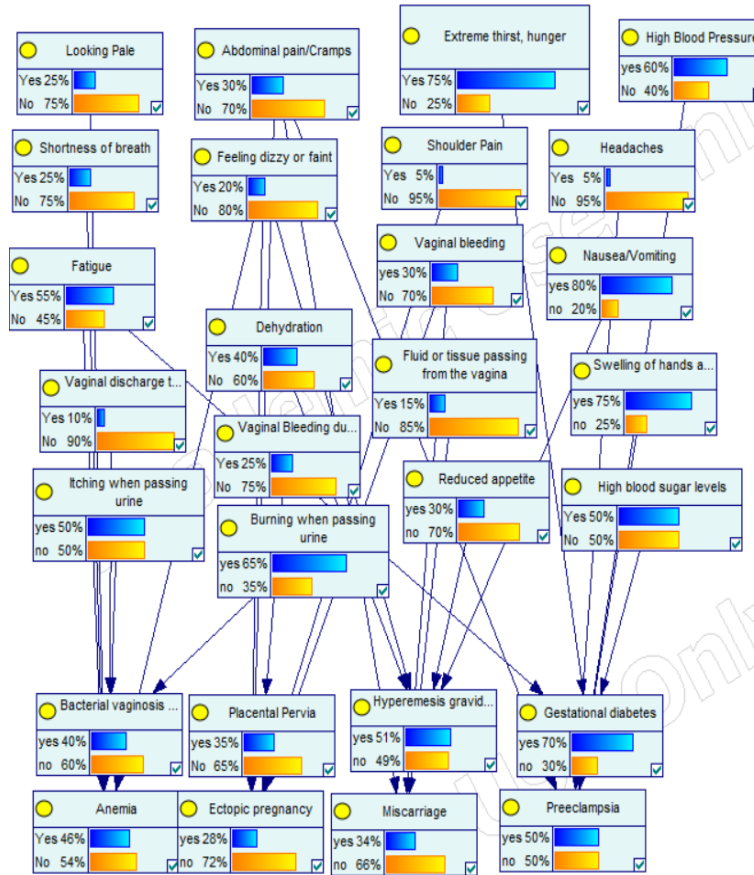


Figure 5.4: Complication Predictor Bayesian Network

Once she was satisfied with the results, the code for Bayesian network was written and integrated into the chatbot.

5.3.2 Integrating Bayesian Network in RASA

The Python library for Bayesian Belief Networks(PyBBN) was used to construct the Bayesian network.

Nodes for symptoms were created and their initial probabilities were defined. Then edges were added between the complication nodes and the symptom nodes indicating which symptoms would be experienced for the particular complication. Lastly, the conditional probabilities of complications given the symptoms were defined.

In order to collect information from the user, RASA form was used. RASA forms are used when a few pieces of information need to be collected from the user. Each piece of information is added to a slot, and once all slots have been filled, a certain action is executed.

For complication prediction, the form was used to ask whether or not the user was experiencing the symptoms that had been defined. The user will be given the options of yes, no or I don't know. This sequence of questions will allow the user's symptoms to be filled in the form of slots. Once all slots are filled, the Bayesian network code executes and the complication will be predicted according to the user symptoms.

The following intents were added, which when detected will activate the form.

```
- intent: complications_prediction|
examples: |
- I dont feel well
- I think I might have a complication
- Is my pregnancy going normally?
- میرے پیٹ میں درد ہے
- مجھے بہت تھکن ہو رہی ہے
- چہرے کی رنگت ختم ہو رہی ہے
- چہرہ پیلا ہو رہا ہے
- شکل پر پیلاہٹ آ رہی ہے
- رنگ عجیب سا ہو گیا ہے
- سانس لینے میں دشواری
- سانس نہیں لیا جا رہا
- سانس کی قلت ہو رہی
- چکر آ رہے ہیں
- قرح سے خون بہہ رہا ہے
- شدید بھوک یا پیاس لگتی ہے
- بے ہوشی سی ہو رہی ہے
- رنگت میں پیلاہٹ آ گئی ہے
- ہر وقت پیاس لگتی رہتی ہے
- ہر وقت بھوک لگتی رہتی ہے
- شوگر بہت زیادہ آئی ہے
- بلٹ شوگر بڑھ گئی ہے
- کندھوں میں درد ہو رہا ہے
- جسم میں درد ہو رہا ہے
- بازو میں درد ہے
- سفید سی چیز نکل رہی ہے
- قرح سے سفید مواد خارج ہوتا ہے
- سر میں درد ہوتا ہے
- متلی یا الٹی ہوتی ہے
- بہت متلی آ رہی ہے
- بہت الٹی آ رہی ہے
- ہر وقت الٹی آتی رہتی ہے
- بائی بلٹ پریشر یا ہائپر ٹینشن ہے
- بلٹ پریشر بہت زیادہ آیا ہے
- بلٹ پریشر بڑھ گیا ہے
- کیا بلٹ پریشر زیادہ آنے سے بچے پر کوئی نقصان ہوگا
- بہت سوجن ہو رہی ہے
- ہاتھ سوج رہے ہیں
- منہ سوج رہا ہے
- چہرہ سوج رہا ہے
- قرح سے مواد خارج ہوتا ہے جس میں ناگوار بو ہے
```

Figure 5.5: Intents to trigger Complication Predictor

5.4 Weekly Messages

Empathetic support is provided to the registered users through weekly text, image, and audio messages. The content of the messages ranges from general information, to nutritional care advice and decision-making advice. The technology used to implement this feature is the Twilio API. Twilio is a developer platform for communications, and through the Twilio Python Helper Library, WhatsApp messages were sent.

5.4.1 Automating Running the Script

Each week, the script containing the code to send messages to each number was automatically executed fixed day and time. This is done using the Windows Task Scheduler- a default Windows Application to execute or perform actions in response of an event based or time based trigger. A .bat file was created that executed the script; within the Scheduler, a basic task was created that got triggered at a fixed time, this task resulted in the running of the .bat file and eventually the messages were sent.

5.4.2 Querying the Database

The script queries the MongoDB database containing the users cellphone number and current week of pregnancy. Using this data, a message is sent to the users phone. Once the message has been sent using Twilio API, the week number is updated and once a woman becomes 43 weeks, her number is removed from the data, as she is likely to have delivered the baby by then.

5.4.3 Twilio API

Twilio's web service APIs allow software developers to make and receive phone calls, send and receive messages on text and WhatsApp, and conduct other communication operations programmatically. A message is created containing either some media (audio or picture) or text, the Twilio business account number that one receives upon registering, and the number you want to send the message to. When the script is executed, due to the `account_sid` and `auth_token`, two identification strings, that are initialized in the code, the code connects to the registered account and the created message is sent successfully using the API.

6. Experiments and Results

6.1 Talking to Hamdard

Using the data, stories, pipeline, and Bayesian Network explained earlier, the RASA chatbot was trained and developed. It was tested thoroughly by trying multiple questions in variety of ways, as well as, by validating data and analyzing the accuracy of the model's predictions and performance through RASA.

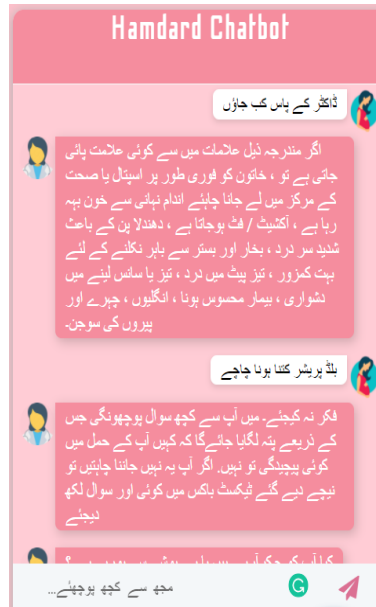


Figure 6.1: Testing Chatbot

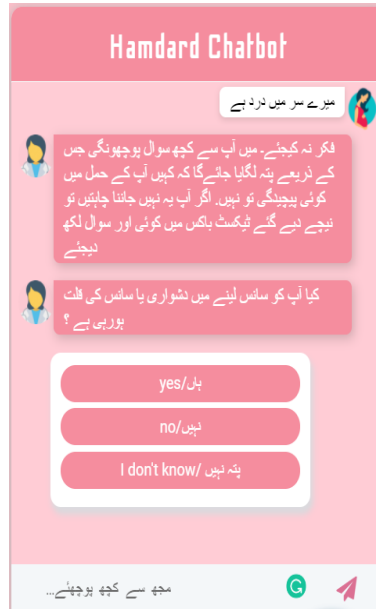


Figure 6.2: Testing Complications Predictor

6.2 Data Validation

Before testing the accuracy of our chatbot, it was essential to check the data itself for errors. This was done by executing `rasa data validate` which checks the outputs the errors present, if any.

6.3 Testing Results

WE will be testing the accuracy of our bot both automatically and manually, in order to better analyze the performance of our bot, Hamdard.

6.3.1 Testing within RASA

The RASA NLU can be tested using two ways; we will test our model using both models.

6.3.1.1 Test/Train Data

One of the techniques to assess a machine learning model is to divide the data into test data and train data, this is done by RASA to ensure same weight-age of intents in both data sets.

```
rasa data split nlu
```

Once the data has been split, the two data sets are used to test the model. The model is trained on the training data and then its detection is verified on the testing data.

```
rasa test nlu -u train_test_split/test_data.md --model models/modelname
```

The results are generated in the form of a histogram and a confusion matrix as shown in Figure 6.3 and 6.4 respectively.

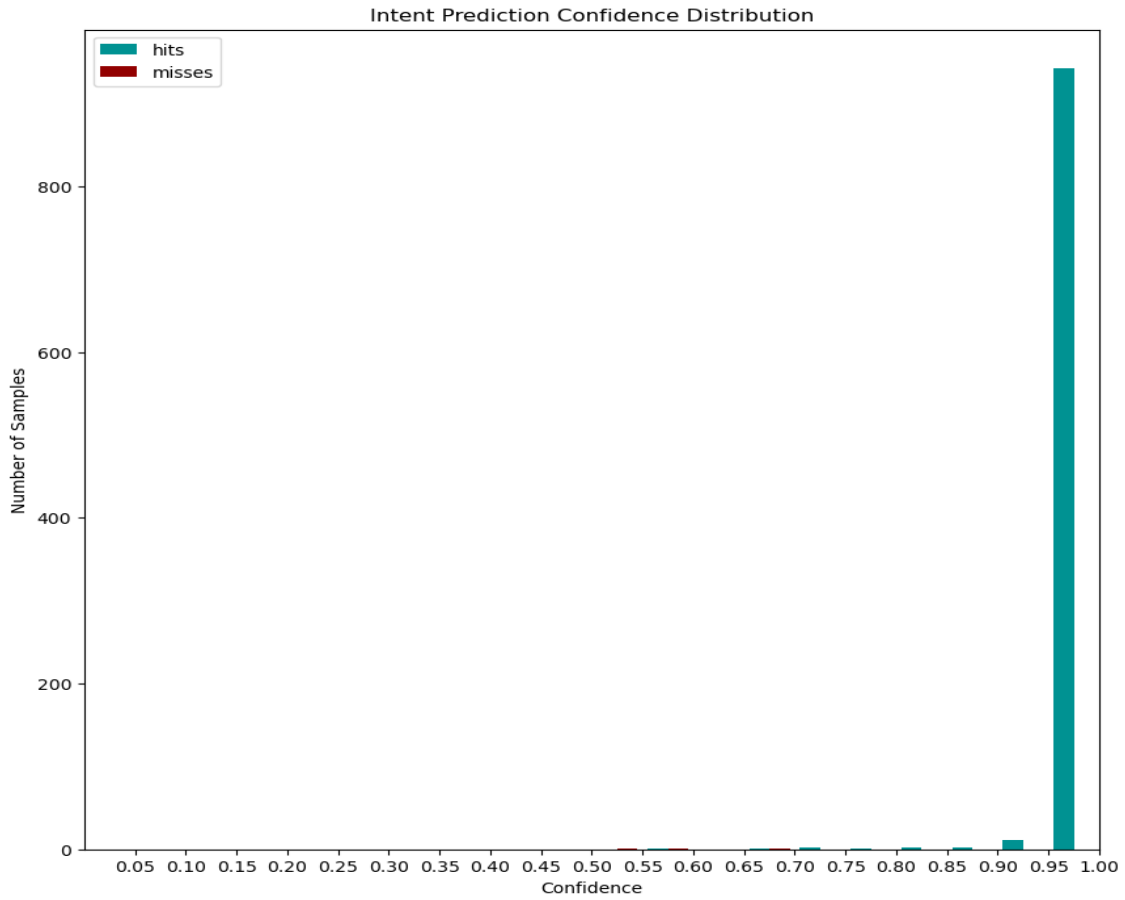


Figure 6.3: Intent Confidence through Test/train data represented as a histogram

The histogram is used to depict the model's confidence in classifying intents. The blue bars show the scores for correct classifications and red bars show incorrect classifications. According to the histogram, the model seems highly accurate with more than 800 samples that were classified correctly at a very high confidence.

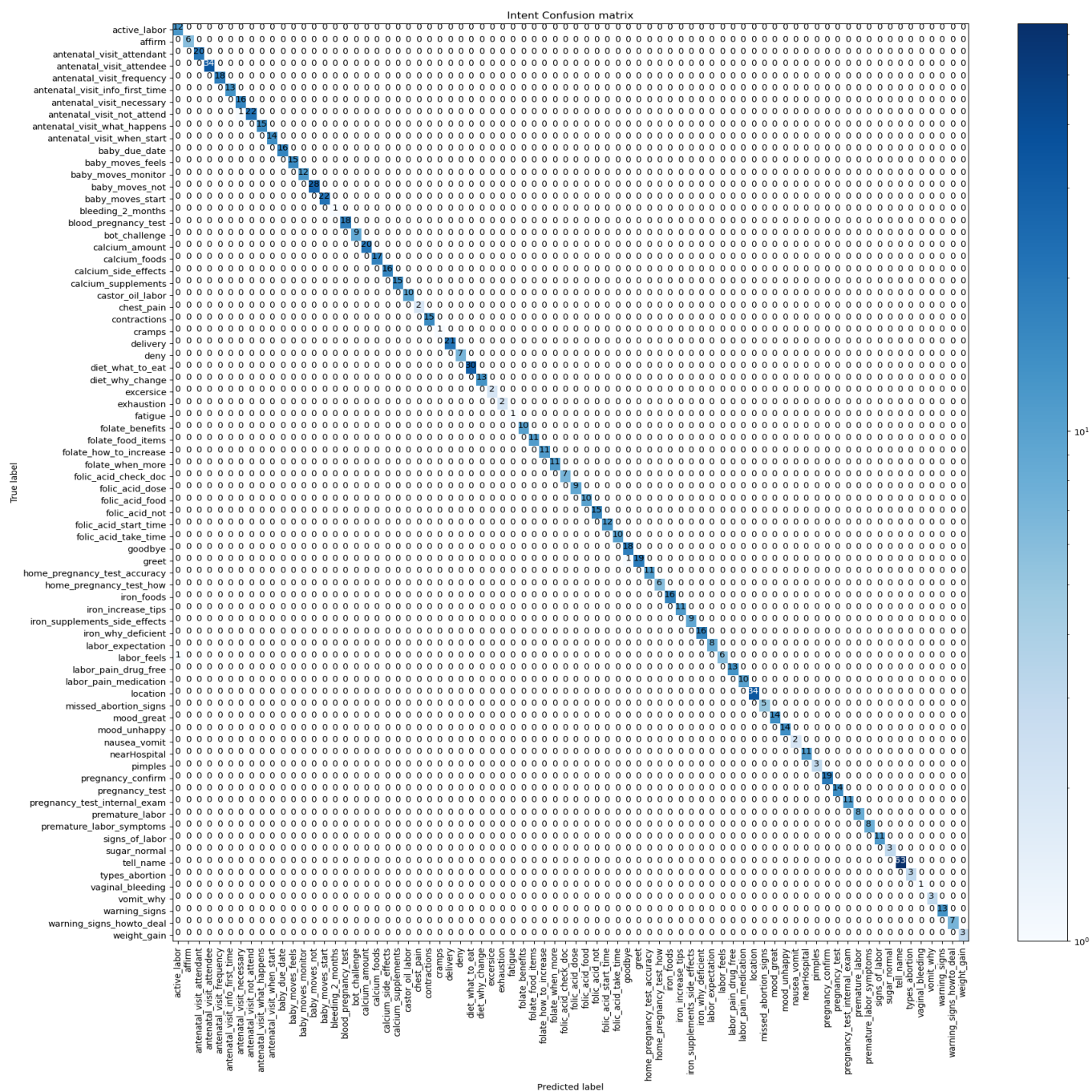


Figure 6.4: Intent Confidence through Test/train data in a confusion matrix

Similarly, the confusion matrix illustrates the number of times the model predicted the correct intent and the number of times it missed which intent for which other, wrongfully. The diagonal shows the number of times the model predicted the intent correctly; since the matrix appears to be a diagonal matrix, it seems to be highly accurate. However, this accuracy seems very idealistic which is why further validation is required.

6.3.1.2 Cross-Validation

An improved technique that allows us to test the model, more vigorously, is through cross-validation. Unlike the previous technique, where our data was divided into half for training and testing, cross validation splits in the entire data set into a specific k (default value = 5) number of groups of equal size. At each fold, one group of data is used for training, while the rest $k-1$ groups are used for testing and evaluating the performance. This happens k times, until each group has been used as for training. Once it has been done k times, the model's performance is validated by the weighted average of all entities. The cross-validation is done by executing the following command.

```
rasa test nlu -u data/nlu.md --config config.yml --cross-validation
```

The results are generated in the form of a histogram and a confusion matrix as in Figure 6.5 and 6.6 respectively.

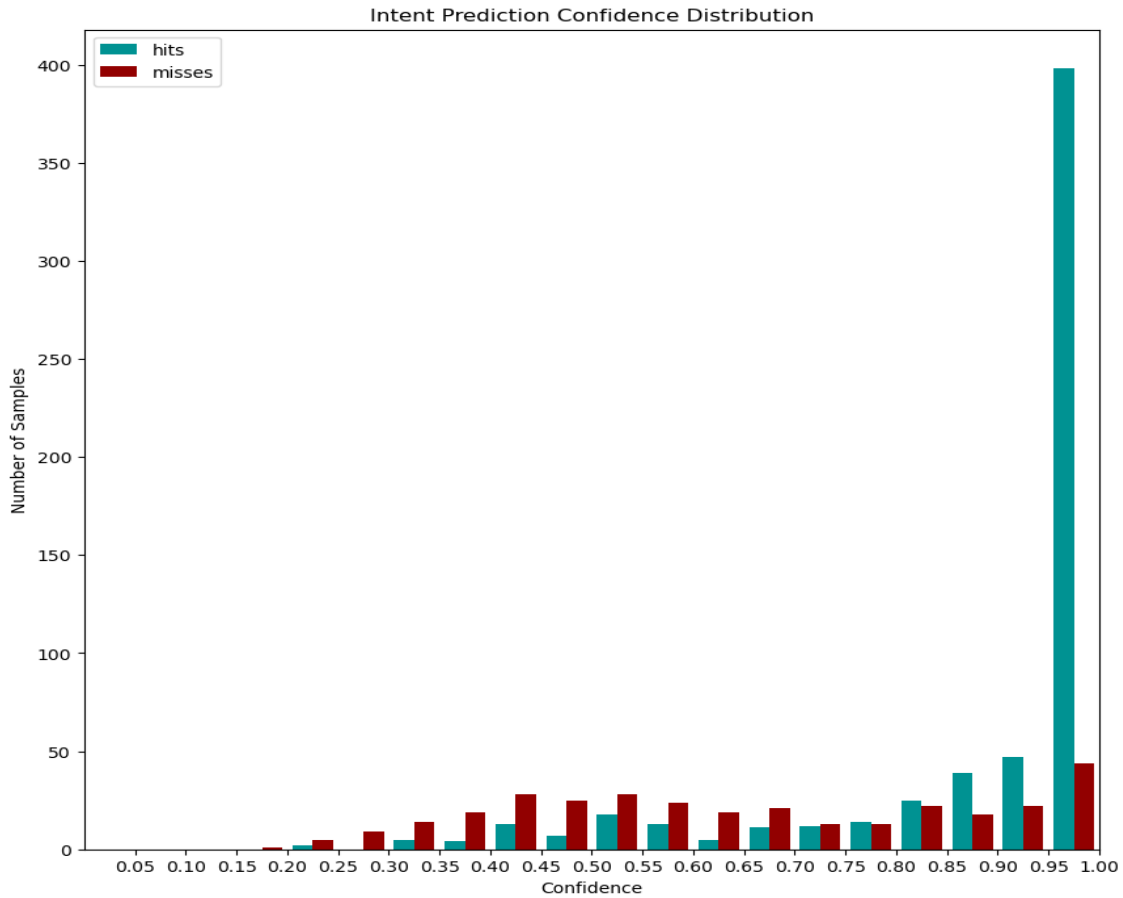


Figure 6.5: Intent Confidence through Cross Validation represented as a histogram

The histogram representation of intent confidence indicates that our model tends to predict the wrong intent when it has a lower confidence, whereas, with intents that are predicted correctly the model tends to have a high confidence about them. To tackle these wrongful predictions, our bot has a fallback policy, which request the user to rephrase the question, in case of low confidence scores. Using this histogram, we decided to keep the fallback threshold to 0.7.

The confusion matrix shows the number of times an intent was detected correctly and how many times it missed for which other intent. In the matrix we see, the maximum number of times our model makes the wrong prediction is 6 times where the intent `iron_increase_tips` is predicted as `iron_foods`. Similarly, `antenatal_visit_attendant` is the most correctly predicted intent with only 3 misses altogether.

These results are very different from the initial technique we used- test/train data split. We can rely on this testing technique more as it provides the average, meaning it basically tests the model k times.

6.3.2 Testing Custom Actions

Testing of the custom actions and the Bayesian Network was done manually. For this purpose, we inputted random symptoms into the bot and when it predicted a complication, we manually checked with our data whether it had predicted it correctly or not. Similarly, to test `action_find_loc` we inputted multiple areas of Karachi and checked whether it provided us with a nearby hospital and if not, whether it gave an apology or not.

6.3.3 Testing using sample User Statements from Survey

Alongside the train/test data split and cross validation, we tested our bot manually as well. We circulated a survey where we asked potential users to list down all possible questions they had during their pregnancies, in roman Urdu. The collected questions acted as our test data set. A total of 75 questions were passed as input to the bot and it was recorded whether the questions were classified as correct intents or not. Since multiple questions could be part of one intent, if the bot identified a few questions correctly and a few incorrectly, it was recorded as **partially correct** and if some intent was predicted incorrectly for all questions it was recorded as **incorrect**.

S.No	Intent	Number of questions asked	Analysis
1	Labor_feels	5	correct
2	Baby_moves_not	7	correct
3	Signs_of_labor	3	correct
4	Baby_moves_start	4	partially correct
5	Pregnancy_test	2	correct
6	Delivery	3	partially correct
7	Antenatal_visit_necessary	1	correct
8	Antenatal_visit_what_happens	3	correct
9	Antenatal_visit_when_start	1	partially correct
10	Baby_moves_not	5	correct
11	Calcium_amount	2	correct
12	Baby_due_date	4	partially correct
13	Blood_pregnancy_test	1	correct
14	Pregnancy_confirm	2	correct
15	Calcium_supplements	2	correct
16	Calcium_side_effects	1	incorrect
17	Baby_moves_feels	4	partially correct
18	Home_pregnancy_test_accuracy	2	partially correct
19	Castor_oil	4	incorrect
20	Folate_benefits	2	correct
21	Folate_food_items	2	correct
22	Diet_why_change	3	incorrect
23	Iron_foods	4	correct
24	Iron_why_deficient	1	correct
25	Labor_pain_drug_free	1	correct
26	Diet_what_to_eat	3	correct
27	Hemoglobin_normal	1	correct
28	Weight_gain	2	partially correct

Table 6.1: User Input Testing

The breakdown of the results is:

Correct	18
Incorrect	3
Partially_correct (Some questions answered correctly, some incorrectly)	5

Table 6.2: Breakdown of User Input Testing

Overall, combining the correct intents and the partially correct intents, the bot was 78-80% accurate, which is for an Urdu chatbot is huge success. However, the performance can be enhanced by adding more training data; this not only allows the bot to cater to other possible questions but also improve the prediction and success of hits of each intent.

7. Conclusion and Future Work

Throughout this project, we aimed to build an assistant that would address women's maternity health issues in Pakistan, whether they be medical or emotional issues. We managed to build an agent by going through a rigorous process of data translation, data verification, creating a web app to host the bot that did not only answer questions, or provide the user with the nearest hospital or center in Karachi, but also predicted complication based on symptoms. We also managed to build the messaging framework where the user received weekly messages. We created the bot using RASA framework, which proved to be the backbone of our assistant, it provided us with NLU support, a template to store training data, integral elements for the pipeline, and even a data validation support. The weekly messages were incorporated into the assistant through Twilio, which provided us with the API to send and receive WhatsApp text and media messages globally.

However, this assistant has potential to be advanced further firstly by adding more data to the data set. By widening the data set, the bot will not only be able to answer more questions but it will also improve the accuracy of its detection. Furthermore, currently the complications predictor has a limited scope, it gives a probability based on the current symptoms only, by including patient medical history, their current week of pregnancy, and other factors, the accuracy of the predictor can also be improved.

This project can also be extended by adding a voice activation feature. By using Speech Recognition and Natural Language Toolkit, users will be able to communicate with the bot via voice. They could ask questions and get answers verbally, it would be like conversing with another person. This would widen the scope of the audience even further and also provide them with an increased emotional and empathetic support. All things considered, Hamdard like any other natural language assistant has scope for development and improvement.

Appendix A. Data

Our chatbot data can be found at [this GitHub link](#) in the RASA files, namely domain.yml, stories.md, nlu.md, and Hosp_data.xlsx The empathetic audio messages can be found [here](#) and the text messages [here](#)

Appendix B. Code

Our code can be found at [this GitHub link](#)

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